

Brandon Jahoor

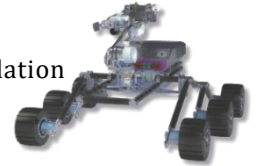


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Candidate for BAsC in Mechatronics Engineering Honours, Minor in Computer Engineering, University of Waterloo
Developing intelligent robotics systems driven by perception and AI!

Skills

- **Robotics & Perception:** ROS2/ROS, Gazebo, Foxglove, OpenCV, kinematics (FK/IK), RealSense, Pure Pursuit
- **Embedded Systems:** Raspberry PI, Nvidia Jetson, STM32(C), ESP32, PLC ladder logic, FPGA/VHDL
- **Software & Tools:** C++, Python, C, Docker, GitHub, Linux/Ubuntu, AI/ML, MATLAB, DSA
- **Design & Prototyping:** SolidWorks(CAD), 3D-printing, electrical/mechanical assembly, testing, validation



Experiences

Robotics Research Assistant – University of Waterloo RoboHub (Sept 2025 - Dec 2025)

- Syncing Xsens Awinda IMUs with 7-axis robotic arm (Emika Panda's) in ROS2 to mirror human arm motion.

Robotics Engineer – University of Waterloo Robotics Team (Co-op) (June 2025 - Aug 2025)

- Developed ROS2 RealSense RGB & depth stream with compression and dynamic resolution selection; added OpenCV processing.
- Modelled/solved 6-wheel rocker-bogie drivetrain forward & inverse kinematics for mars rover running on Nvidia Jetson.

Robotics Engineer – BHF Robotics (Co-op) (May 2025 - June 2025)

- Implemented ROS2 autodrive routine; trained deep learning AI model for autonomous farming robot's weed/crop detection.
- Diagnosed electrical issues and restored robotic controls; contributed to part design & simulation for prototyping/manufacturing.

Mechatronics Engineer – RAB Design Lighting (Co-op) (Sept 2024 - Dec 2024)

- Validated and tested lighting controls app; authored app manual and training materials ensuring successful app launch.
- Regularly used SolidWorks/AutoCAD for design/documentation; produced engineering deliverables & process improvement tools.

Projects

Autonomous Rover Simulation – Gazebo, ROS2, Python, Navigation, AI

- Developed a Gazebo rover simulation with ROS2 navigation stack with obstacle avoidance for autonomous goal-seeking movement.
- Added AI-based plant detection node using onboard camera feed and image classification model.

AI Object Detection – AI, ROS2, Python, RealSense, OpenCV

- Developed ROS2 RealSense camera package with compression for efficient RGB-D streaming and local AI inference.
- Integrated Hugging Face Transformers (OWL-ViT) for zero-shot, text-prompt object recognition with real-time OpenCV visualization.

WATonomous – Docker, Foxglove, C++, LiDAR, A*

- Implementing ROS2 navigation stack with A* global path planning & Pure-Pursuit control within Docker for autonomous navigation.
- Using Foxglove to visualize LiDAR-based cost maps, path planning, and odometry in real-time.

SWARM Robots – Linux/Ubuntu, Docker, Gazebo, ROS2 Assembly, ESP32

- Assembled and programmed a fleet of HeRo 2.0 swarm robots from scratch, including SMD soldering & ESP32 WiFi communication.
- Deployed ROS2 and Gazebo containers in Docker to simulate swarm coordination.

3-DOF Robotic Arm – ROS2, OpenCV, SolidWorks, PWM, 3D Printing, Raspberry PI

- Designed & 3D-printed robotic arm using SolidWorks; implemented PWM motor control on Raspberry Pi running ROS2.
- Programmed colour-based token sorting using OpenCV vision and HSV filtering for real-time classification.



For more information, please visit my website portfolio: bjahoor.github.io